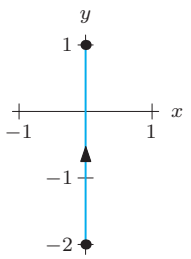
Exercises and Problems for Section 17.1

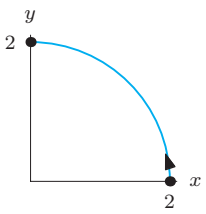
Exercises

In Exercises 1–6, find a parameterization for the curve shown.

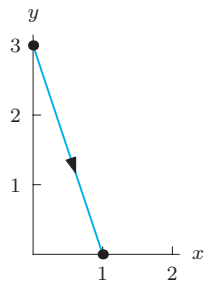
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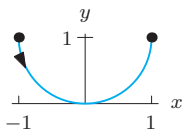
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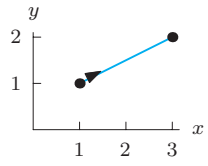
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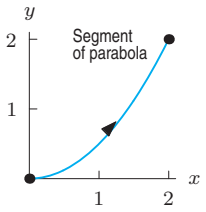
4.



5.



6.



In Exercises 7–17, find parametric equations for the line.

7. The line in the direction of the vector $\vec{i} - \vec{k}$ and through the point $(0, 1, 0)$.

8. The line in the direction of the vector $\vec{i} + 2\vec{j} - \vec{k}$ and through the point $(3, 0, -4)$.

9. The line parallel to the z-axis passing through the point $(1, 0, 0)$.

10. The line in the direction of the vector $5\vec{j} + 2\vec{k}$ and through the point $(5, -1, 1)$.

11. The line in the direction of the vector $3\vec{i} - 3\vec{j} + \vec{k}$ and through the point $(1, 2, 3)$.

12. The line in the direction of the vector $2\vec{i} + 2\vec{j} - 3\vec{k}$ and through the point $(-3, 4, -2)$.

13. The line through $(-3, -2, 1)$ and $(-1, -3, -1)$.

14. The line through the points $(1, 5, 2)$ and $(5, 0, -1)$.

15. The line through the points $(2, 3, -1)$ and $(5, 2, 0)$.

16. The line through $(3, -2, 2)$ and intersecting the y-axis at $y = 2$.

17. The line intersecting the x-axis at $x = 3$ and the z-axis at $z = -5$.

In Exercises 18–34, find a parameterization for the curve.

18. A line segment between $(2, 1, 3)$ and $(4, 3, 2)$.

19. A circle of radius 3 centered on the z-axis and lying in the plane $z = 5$.

20. A line perpendicular to the plane $z = 2x - 3y + 7$ and through the point $(1, 1, 6)$.

21. The circle of radius 2 in the xy-plane, centered at the origin, clockwise.

22. The circle of radius 2 parallel to the xy-plane, centered at the point $(0, 0, 1)$, and traversed counterclockwise when viewed from below.

23. The circle of radius 2 in the xz-plane, centered at the origin.

24. The circle of radius 3 parallel to the xy-plane, centered at the point $(0, 0, 2)$.

25. The circle of radius 3 in the yz-plane, centered at the point $(0, 0, 2)$.

26. The circle of radius 5 parallel to the yz-plane, centered at the point $(-1, 0, -2)$.

27. The curve $x = y^2$ in the xy-plane.

28. The curve $y = x^3$ in the xy-plane.

29. The curve $x = -3z^2$ in the xz-plane.

30. The curve in which the plane $z = 2$ cuts the surface $z = \sqrt{x^2 + y^2}$.

31. The curve $y = 4 - 5x^4$ through the point $(0, 4, 4)$, parallel to the xy-plane.

32. The ellipse of major diameter 5 parallel to the y-axis and minor diameter 2 parallel to the z-axis, centered at $(0, 1, -2)$.

33. The ellipse of major diameter 6 along the x-axis and minor diameter 4 along the y-axis, centered at the origin.

34. The ellipse of major diameter 3 parallel to the x-axis and minor diameter 2 parallel to the z-axis, centered at $(0, 1, -2)$.

In Exercises 35–42, find a parametric equation for the curve segment. (There are many possible answers.)

35. Line from $(-1, 2, -3)$ to $(2, 2, 2)$.

36. Line from $P_0 = (-1, -3)$ to $P_1 = (5, 2)$.

37. Line from $P_0 = (1, -3, 2)$ to $P_1 = (4, 1, -3)$.

38. Semicircle from $(0, 0, 5)$ to $(0, 0, -5)$ in the yz-plane with $y \geq 0$.

39. Semicircle from $(1, 0, 0)$ to $(-1, 0, 0)$ in the xy-plane with $y \geq 0$.

40. Graph of $y = \sqrt{x}$ from $(1, 1)$ to $(16, 4)$.

41. Arc of a circle of radius 5 from $P = (0, 0)$ to $Q = (10, 0)$.

42. Quarter-ellipse from $(4, 0, 3)$ to $(0, -3, 3)$ in the plane $z = 3$.

Problems

In Problems 43–47, parameterize the line through $P = (2, 5)$ and $Q = (12, 9)$ so that the points P and Q correspond to the given parameter values.

43. $t = 0$ and 1

44. $t = 0$ and 5

45. $t = 20$ and 30

46. $t = 10$ and 11

47. $t = 0$ and -1

48. At the point where $t = -1$, find an equation for the plane perpendicular to the line

$$x = 5 - 3t, \quad y = 5t - 7, \quad \frac{z}{t} = 6.$$

49. Determine whether the following line is parallel to the plane $2x - 3y + 5z = 5$:

$$x = 5 + 7t, \quad y = 4 + 3t, \quad z = -3 - 2t.$$

50. Show that the equations $x = 3 + t$, $y = 2t$, $z = 1 - t$ satisfy the equations $x + y + 3z = 6$ and $x - y - z = 2$. What does this tell you about the curve parameterized by these equations?

51. **(a)** Explain why the line of intersection of two planes must be parallel to the cross product of a normal vector to the first plane and a normal vector to the second.
(b) Find a vector parallel to the line of intersection of the two planes $x + 2y - 3z = 7$ and $3x - y + z = 0$.
(c) Find parametric equations for the line in part (b).

52. Find an equation for the plane containing the point $(2, 3, 4)$ and the line $x = 1 + 2t, y = 3 - t, z = 4 + t$.

53. **(a)** Find an equation for the line perpendicular to the plane $2x - 3y = z$ and through the point $(1, 3, 7)$.
(b) Where does the line cut the plane?
(c) What is the distance between the point $(1, 3, 7)$ and the plane?

54. Consider two points P_0 and P_1 in 3-space.

(a) Show that the line segment from P_0 to P_1 can be parametrized by

$$\vec{r}\,(t) = (1 - t)\overrightarrow{OP_0} + t\overrightarrow{OP_1}, \quad 0 \le t \le 1.$$

(b) What is represented by the parametric equation

$$\vec{r}\,(t) = t\overrightarrow{OP_0} + (1 - t)\overrightarrow{OP_1}, \quad 0 \le t \le 1?$$

55. **(a)** Find a vector parallel to the line of intersection of the planes $2x - y - 3z = 0$ and $x + y + z = 1$.
(b) Show that the point $(1, -1, 1)$ lies on both planes.
(c) Find parametric equations for the line of intersection.

56. Find the intersection of the line $x = 5 + 7t$, $y = 4 + 3t$, $z = -3 - 2t$ and the plane $2x - 3y + 5z = -7$.

In Problems 57–59 two parameterized lines are given. Are they the same line?

57. $\vec{r}_1(t) = (5 - 3t)\vec{i} + 2t\vec{j} + (7 + t)\vec{k}$
$\vec{r}_2(t) = (5 - 6t)\vec{i} + 4t\vec{j} + (7 + 3t)\vec{k}$

58. $\vec{r}_1(t) = (5 - 3t)\vec{i} + (1 + t)\vec{j} + 2t\vec{k}$
$\vec{r}_2(t) = (2 + 6t)\vec{i} + (2 - 2t)\vec{j} + (2 - 4t)\vec{k}$

59. $\vec{r}_1(t) = (5 - 3t)\vec{i} + (1 + t)\vec{j} + 2t\vec{k}$
$\vec{r}_2(t) = (2 + 6t)\vec{i} + (2 - 2t)\vec{j} + (3 - 4t)\vec{k}$

60. If it exists, find the value of c for which the lines $l(t) = (c+t, 1+t, 5+t)$ and $m(s) = (s, 1-s, 3+s)$ intersect.

61. **(a)** Where does the line $\vec{r} = 2\vec{i} + 5\vec{j} + t(3\vec{i} + \vec{j} + 2\vec{k})$ cut the plane $x + y + z = 1$?
(b) Find a vector perpendicular to the line and lying in the plane.
(c) Find an equation for the line that passes through the point of intersection of the line and plane, is perpendicular to the line, and lies in the plane.

In Problems 62–65, find parametric equations for the line.

62. The line of intersection of the planes $x - y + z = 3$ and $2x + y - z = 5$.

63. The line of intersection of the planes $x + y + z = 3$ and $x - y + 2z = 2$.

64. The line perpendicular to the surface $z = x^2 + y^2$ at the point $(1, 2, 5)$.

65. The line through the point $(-4, 2, 3)$ and parallel to a line in the yz-plane which makes a 45° angle with the positive y-axis and the positive z-axis.

66. Is the point $(-3, -4, 2)$ visible from the point $(4, 5, 0)$ if there is an opaque ball of radius 1 centered at the origin?

67. Two particles are traveling through space. At time t the first particle is at the point $(-1 + t, 4 - t, -1 + 2t)$ and the second particle is at $(-7 + 2t, -6 + 2t, -1 + t)$.

(a) Describe the two paths in words.
(b) Do the two particles collide? If so, when and where?
(c) Do the paths of the two particles cross? If so, where?

68. For $t > 0$, a particle moves along the curve $x = a + b\sin kt$, $y = a + b\cos kt$, where a, b, k are positive constants.

(a) Describe the motion in words.
(b) What is the effect on the curve of the following changes?
(i) Increasing b
(ii) Increasing a
(iii) Increasing k
(iv) Setting a and b equal

69. In the Atlantic Ocean off the coast of Newfoundland, Canada, the temperature and salinity (saltiness) vary throughout the year. Figure 17.9 shows a parametric curve giving the average temperature, T (in °C) and salinity (in grams of salt per kg of water) for t in months, with $t = 1$ corresponding to mid-January.[1]

(a) Why does the parameterized curve form a loop?
(b) When is the water temperature highest?
(c) When is the water saltiest?
(d) Estimate dT/dt at $t = 6$, and give the units. What is the meaning of your answer for seawater?

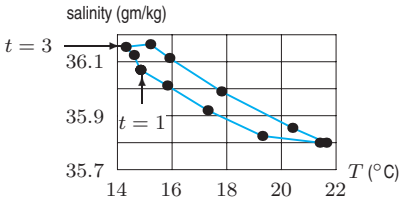


Figure 17.9

70. The concentration (in micrograms/m^3) of a pollutant at the point (x, y, z) is given, for x, y, z in meters, by

$$c(x, y, z) = e^{-(x^2+y^2+z^2)}.$$

(a) A particle at point $(0, 1, 0)$ moves in the direction of the point $(-2, 2, -2)$. What is the rate at which the concentration is changing with respect to distance at the moment the particle leaves the point $(0, 1, 0)$?
(b) Now suppose the particle is moving along the path, with time, t, in seconds, given by

$$x = 1-t^2, \quad y = t, \quad z = 1-t^2 \quad \text{for } -\infty < t < \infty.$$

When is the concentration a maximum?

71. A light shines on the helix of Example 3 on page 918 from far down each axis. Sketch the shadow the helix casts on each of the coordinate planes: xy, xz, and yz.

72. For a positive constant a and $t \geq 0$, the parametric equations I-V represent the curves described in (a)-(e). Match each description (a)-(e) with its parametric equations and write an equation involving only x and y for the curve.

(a) Line through the origin.
(b) Line not through the origin.
(c) Hyperbola opening along x-axis.
(d) Circle traversed clockwise.
(e) Circle traversed counterclockwise.

I. $x = a\sin t, y = a\cos t$ II. $x = a\sin t, y = a\sin t$
III. $x = a\cos t, y = a\sin t$ IV. $x = a\cos^2 t, y = a\sin^2 t$
V. $x = a/\cos t, y = a\tan t$

73. **(a)** Find a parametric equation for the line through the point $(2, 1, 3)$ and in the direction of $a\vec{i} + b\vec{j} + c\vec{k}$.
(b) Find conditions on a, b, c so that the line you found in part (a) goes through the origin. Give a reason for your answer.

74. Consider the line $x = 5 - 2t, y = 3 + 7t, z = 4t$ and the plane $ax + by + cz = d$. All the following questions have many possible answers. Find values of a, b, c, d such that:

(a) The plane is perpendicular to the line.
(b) The plane is perpendicular to the line and through the point $(5, 3, 0)$.
(c) The line lies in the plane.

75. Explain the significance of the constants $\alpha > 0$ and $\beta > 0$ in the family of helices given by $\vec{r} = \alpha\cos t\vec{i} + \alpha\sin t\vec{j} + \beta t\vec{k}$.

76. Find parametric equations of the line passing through the points $(1, 2, 3)$, $(3, 5, 7)$ and calculate the shortest distance from the line to the origin.

77. A line has equation $\vec{r} = \vec{a} + t\vec{b}$ where $\vec{r} = x\vec{i} + y\vec{j} + z\vec{k}$ and $\vec{a}$ and $\vec{b}$ are constant vectors such that $\vec{a} \neq \vec{0}, \vec{b} \neq \vec{0}$, $\vec{b}$ not parallel or perpendicular to $\vec{a}$. For each of the planes (a)–(c), pick the equation (i)–(ix) which represents it. Explain your choice.

(a) A plane perpendicular to the line and through the origin.
(b) A plane perpendicular to the line and not through the origin.
(c) A plane containing the line.

(i) $\vec{a} \cdot \vec{r} = ||\vec{b}||$ (ii) $\vec{b} \cdot \vec{r} = ||\vec{a}||$
(iii) $\vec{a} \cdot \vec{r} = \vec{b} \cdot \vec{r}$ (iv) $(\vec{a} \times \vec{b}) \cdot (\vec{r} - \vec{a}) = 0$
(v) $\vec{r} - \vec{a} = \vec{b}$ (vi) $\vec{a} \cdot \vec{r} = 0$
(vii) $\vec{b} \cdot \vec{r} = 0$ (viii) $\vec{a} + \vec{r} = \vec{b}$
(ix) $(\vec{a} \times \vec{b}) \cdot (\vec{r} - \vec{b}) = ||\vec{a}||$

78. **(a)** Find a parametric equation for the line through the point $(1, 5, 2)$ and in the direction of the vector $2\vec{i} + 3\vec{j} - \vec{k}$.
(b) By minimizing the square of the distance from a point on the line to the origin, find the exact point on the line which is closest to the origin.

79. A plane from Denver, Colorado, (altitude 1650 meters) flies to Bismark, North Dakota (altitude 550 meters). It travels at 650 km/hour along a horizontal line at 8,000 meters above the line joining Denver and Bismark. Bismark is about 850 km in the direction 60° north of east from Denver. Find parametric equations describing the plane's motion. Assume the origin is at sea level beneath Denver, that the x-axis points east and the y-axis points north, and that the earth is flat. Measure distances in kilometers and time in hours.

[1] Based on http://www.vub.ac.be. Accessed Nov 2011.

80. The plane $x+3y-2z=6$ is colored blue and the plane $2x+y+z=3$ is colored yellow. The planes intersect in a line, which is colored green. You are at the point $P=(1,-2,-1)$.

(a) You look in the direction $\vec{v}=\vec{i}+2\vec{j}+\vec{k}$. Do you see the blue plane or the yellow plane?

(b) In what direction(s) are you looking directly at the green line?

(c) In what direction(s) should you look to see the yellow plane? The blue plane?

81. The vector $\vec{n}$ is perpendicular to the plane P_1. The vector $\vec{v}$ is parallel to the line L.

(a) If $\vec{n}\cdot\vec{v}=0$, what does this tell you about the directions of P_1 and L? (Are they parallel? Perpendicular? Or is it impossible to tell?)

(b) Suppose $\vec{n}\times\vec{v}\neq\vec{0}$. The plane P_2 has normal $\vec{n}\times\vec{v}$. What can you say about the directions of

(i) P_1 and P_2? (ii) L and P_2?

82. Figure 17.10 shows the parametric curve $x=x(t), y=y(t)$ for $a\le t\le b$.

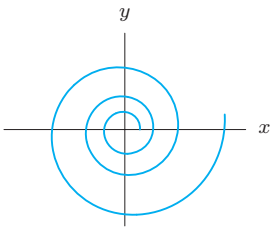
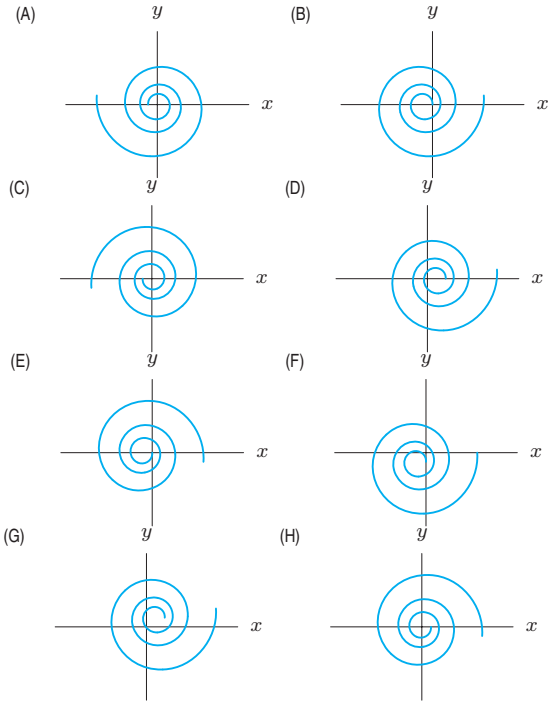


Figure 17.10

(a) Match a graph to each of the parametric curves given, for the same t values, by

(i) $(-x(t),-y(t))$ (ii) $(-x(t),y(t))$
(iii) $(x(t)+1,y(t))$ (iv) $(x(t)+1,y(t)+1)$



(b) Which of the following could be the formulas for the functions $x(t), y(t)$?

(i) $x=10\cos t \quad y=10\sin t$
(ii) $x=(10+8t)\cos t \quad y=(10+8t)\sin t$
(iii) $x=e^{t^2/200}\cos t \quad y=e^{t^2/200}\sin t$
(iv) $x=(10-8t)\cos t \quad y=(10-8t)\sin t$
(v) $x=10\cos(t^2+t) \quad y=10\sin(t^2+t)$

Strengthen Your Understanding

In Problems 83–84, explain what is wrong with the statement.

83. The curve parameterized by $\vec{r}_1(t)=\vec{r}(t-2)$, defined for all t, is a shift in the $\vec{i}$-direction of the curve parameterized by $\vec{r}(t)$.

84. All points of the curve $\vec{r}(t)=R\cos t\vec{i}+R\sin t\vec{j}+t\vec{k}$ are the same distance, R, from the origin.

In Problems 85–87, give an example of:

85. Parameterizations of two different circles that have the same center and equal radii.

86. Parameterizations of two different lines that intersect at the point $(1,2,3)$.

87. A parameterization of the line $x=t, y=2t, z=3+4t$ that is not given by linear functions.

Are the statements in Problems 88–99 true or false? Give reasons for your answer.

88. The parametric curve $x=3t+2, y=-2t$ for $0\le t\le 5$ passes through the origin.

89. The parametric curve $x=t^2, y=t^4$ for $0\le t\le 1$ is a parabola.

90. A parametric curve $x=g(t), y=h(t)$ for $a\le t\le b$ is always the graph of a function $y=f(x)$.

91. The parametric curve $x=(3t+2)^2, y=(3t+2)^2-1$ for $0\le t\le 3$ is a line.

92. The parametric curve $x = -\sin t$, $y = -\cos t$ for $0 \le t \le 2\pi$ traces out a unit circle counterclockwise as t increases.

93. A parameterization of the graph of $y = \ln x$ for $x > 0$ is given by $x = e^t, y = t$ for $-\infty < t < \infty$.

94. Both $x = -t + 1, y = 2t$ and $x = 2s, y = -4s + 2$ describe the same line.

95. The line of intersection of the two planes $z = x + y$ and $z = 1 - x - y$ can be parameterized by $x = t, y = \frac{1}{2} - t, z = \frac{1}{2}$.

96. The two lines given by $x = t, y = 2 + t, z = 3 + t$ and $x = 2s, y = 1 - s, z = s$ do not intersect.

97. The line parameterized by $x = 1, y = 2t, z = 3 + t$ is parallel to the x-axis.

98. The equation $\vec{r}\,(t) = 3t\vec{i} + (6t + 1)\vec{j}$ parameterizes a line.

99. The lines parameterized by $\vec{r}_1(t) = t\vec{i} + (-2t + 1)\vec{j}$ and $\vec{r}_2(t) = (2t + 5)\vec{i} + (-t)\vec{j}$ are parallel.

17.2 MOTION, VELOCITY, AND ACCELERATION

In this section we see how to find the vector quantities of velocity and acceleration from a parametric equation for the motion of an object.

The Velocity Vector

The velocity of a moving particle can be represented by a vector with the following properties:

The **velocity vector** of a moving object is a vector $\vec{v}$ such that:
- The magnitude of $\vec{v}$ is the speed of the object.
- The direction of $\vec{v}$ is the direction of motion.

Thus, the speed of the object is $\|\vec{v}\,\|$ and the velocity vector is tangent to the object's path.

Example 1 A child is sitting on a Ferris wheel of diameter 10 meters, making one revolution every 2 minutes. Find the speed of the child and draw velocity vectors at two different times.

Solution The child moves at a constant speed around a circle of radius 5 meters, completing one revolution every 2 minutes. One revolution around a circle of radius 5 is a distance of 10π, so the child's speed is $10\pi/2 = 5\pi \approx 15.7$ m/min. Hence, the magnitude of the velocity vector is 15.7 m/min. The direction of motion is tangent to the circle, and hence perpendicular to the radius at that point. Figure 17.11 shows the direction of the vector at two different times.

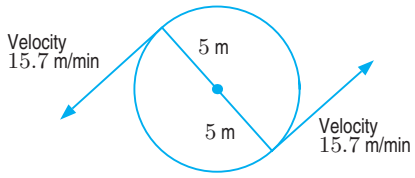


Figure 17.11: Velocity vectors of a child on a Ferris wheel (Note that vectors would be in opposite direction if viewed from the other side)

Computing the Velocity

We find the velocity, as in one-variable calculus, by taking a limit. If the position vector of the particle is $\vec{r}\,(t)$ at time t, then the displacement vector between its positions at times t and $t + \Delta t$ is $\Delta\vec{r} = \vec{r}\,(t + \Delta t) - \vec{r}\,(t)$. (See Figure 17.12.) Over this interval,

$$\text{Average velocity} = \frac{\Delta\vec{r}}{\Delta t}.$$